

project-2

Student Grade System

Project Title: Student Grade System in Java

Project Objective:

The objective of this project is to develop a simple console-based application to calculate student marks, average, and grades. This project helps in understanding important concepts of Java Programming such as loops, conditions, exception handling, and user input handling.

Project Description:

The Student Grade System is a Java application where the user can enter details of multiple students. For each student, the program takes the student's name, number of subjects, and marks for each subject. It then calculates the **total marks**, **average percentage**, and assigns a **grade** based on the average.

The grading system is:

- **A** → 90 and above
- **B** → 80 to 89
- **C** → 70 to 79
- **D** → 60 to 69
- **E** → 50 to 59
- **Fail** → Below 50

This system can process multiple students in one execution.

Technologies Used:

- Java Programming
- Scanner Class

Concepts Used:

- **for** loop

- `do-while` loop
- `If-else if-else` statements
- `try-catch` exception handling
- User input handling
- Average calculation
- Grade assignment

Code Explanation:

The program starts by taking the total number of students. A `for` loop is used to process each student one by one. For every student, the program asks for their name and the number of subjects. A `do-while` loop is used to input marks for each subject. The marks are validated to ensure they are between 0 and 100. If invalid marks are entered, the user is asked again. The total marks are calculated by summing all subject marks. The average is calculated by dividing total marks by number of subjects. Based on the average, grades are assigned using `if-else` conditions. A `try-catch` block handles invalid inputs.

Formula Used:

Average Formula:

Average = Total Marks / Number of Subjects

Advantages:

- Easy to use
- Supports multiple students
- Calculates marks and grades quickly
- Prevents invalid marks input
- Beginner-friendly project

```
Welcome to student grade system
Enter no of students:
2
Enter student name:
akhila
Hello akhila, check your marks and grade below
Enter no of subjects: 3
Enter sub 1 marks(0 to 100): abc
Invalid input! Please enter a number.
Enter sub 1 marks(0 to 100): 100
Enter sub 2 marks(0 to 100): 100
Enter sub 3 marks(0 to 100): 90
Student Marks Report
Total Marks: 290
Average Marks:96.67% Your Grade is: A
Enter student name:
tulasi sravani
Hello tulasi sravani, check your marks and grade below
Enter no of subjects: 4
Enter sub 1 marks(0 to 100): 40
Enter sub 2 marks(0 to 100): 70
Enter sub 3 marks(0 to 100): abc
Invalid input! Please enter a number.
Enter sub 3 marks(0 to 100): 20
Enter sub 4 marks(0 to 100): 79
Student Marks Report
Total Marks: 209
Average Marks:52.25% Your Grade is: E
Thank you for visit.
```

Conclusion:

This Student Grade System project is a beginner-friendly Java application that demonstrates practical use of loops, conditions, input validation, and exception handling. It is useful for understanding how student result processing works and improves problem-solving skills in Java Programming.